Yoel Kim

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Research Interests

- Software Engineering → software verification; specification mining; embedded software.
- Formal Methods → abstraction; automata learning; model checking.
- Programming Languages → program analysis; program synthesis.

Education

Ph.D. in Computer Science and Engineering, Kyungpook National University — *Mar 2023 – Feb 2027 (Expected)*
- Advisor: Yunja Choi
- Dissertation: End-to-End Active Learning of Symbolic Automata for Stateful C Programs

M.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2021 – Feb 2023*
- Advisor: Yunja Choi
- Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification

B.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2017 – Feb 2021*
- Track: Global Software Convergence
- GPA: 3.95/4.3, Rank: 1/13

Ongoing Work

1. AutoCL: End-to-End Active Learning of Symbolic Automata for Stateful C Programs
   Yoel Kim, Yong-Hwan Jeong, and Yunja Choi
   (In preparation) **TACAS'27**: *International Conference on Tools and Algorithms for the Construction and Analysis of Systems* (Regular Tool Papers Track)
2. Learning Functional Specifications from Stateful Control Programs through Active Model Learning
   Yunja Choi and Yoel Kim
   (Submitted) *IEEE Access*, 2026

International Conferences

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper
   Yoel Kim and Yunja Choi
   FSE'26: *ACM International Conference on the Foundations of Software Engineering*
2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software
   Yoel Kim and Yunja Choi
   FSE'24: *ACM International Conference on the Foundations of Software Engineering*

Domestic Conferences

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KCSE'26: *Korea Conference on Software Engineering* ** Short Paper Award*
2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation
   Yoel Kim and Yunja Choi
   KCSE'23: *Korea Conference on Software Engineering* ** Best Short Paper Award*
3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis
   Yoel Kim and Yunja Choi
   KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain
   Yoel Kim and Yunja Choi
   KCSE'21: *Korea Conference on Software Engineering*

Domestic Journals

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation (Extended version of the KCSE'26 paper)
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KIPS Transactions on Software and Data Engineering, 2026

Selected Talks

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Research Paper Presentation. FSE'26. *Concordia University, Montreal, Canada. Jul 7, 2026.*
2. AutoCL: Active Learning of Symbolic Automata via Dynamic Symbolic Mapper. Milestone Talk. 10th STAAR Workshop. *Ulsan, Korea. Feb 3, 2026.*
3. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. Milestone Talk. 9th STAAR Workshop. *Korea University, Seoul, Korea. Jul 29, 2025.*
4. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCC'25. *Jeju, Korea. Jul 4, 2025.*
5. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*
6. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research Paper Presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*

Experience

Teaching Assistant, Kyungpook National University *Mar 2021 – Jun 2024*
- ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).
- COMP0224: Software Design (Fall 2021, 2022).
- COMP0216: Data Structure Applications (Spring 2021).

Undergraduate Tutor, Kyungpook National University *Mar 2018 – Dec 2020*
- COMP0322: Database Management Systems (Fall 2020).
- CLTR266: Software Computational Thinking (Spring 2018, 2019).
- CLTR268: Python Programming (Winter 2018).
- COMP0216: Data Structure Applications (Fall 2018).

Undergraduate Research Intern, Software Safety Engineering Lab *Jul 2019 – Dec 2019; Sep 2020 – Feb 2021*
- Supervisor: Yunja Choi
- Developed a Java-based GUI tool to support software testing activities for Hyundai Motor.

Intern, JS System, Daegu *Dec 2019 – Feb 2020*
- Assisted with testing an FPGA-based embedded board.

Undergraduate Research Intern, Intelligent Network Lab *Oct 2018 – Dec 2018*
- Supervisor: Dongkyun Kim